Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 12 · SLO 1 — Solved Worksheet

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Unit â€“ 2 : Session â€“ 12 : SLO â€“ 1 | Implementation of Doubly Linked Lists â€” Concepts I

I. Fill in the Blanks

• A doubly linked list node typically contains data, a pointer to the next node, and a pointer to the PREVIOUS node.
• The structure of a doubly linked list node in C often uses the keyword STRUCT.
• In C, to create a node dynamically, we use the function MALLOC().
• The first node of a doubly linked list is called the HEAD (START).
• To insert a node at the beginning, the new node's next pointer should point to the OLD HEAD (first) node.
• When inserting at the beginning, the previous pointer of the old head should point to the NEW node.
• To insert a node at the end, we must traverse the list until we find the node whose next is NULL.
• While inserting at the end, the new node's previous pointer should point to the LAST (current tail) node.
• To delete the first node, we move the head pointer to the SECOND (next) node.
• When deleting the first node, the new head's previous pointer must be set to NULL.